

MindMitra

System Architecture

COMPASS Response Generation × MEMOIR Memory — Unified Design

One architecture. No flags. No fallbacks. Production-grade.

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Part 1 — System Overview

1. What MindMitra Is

MindMitra is a mental health companion AI built for India. Its users are experiencing anxiety, loneliness, grief, burnout, and suicidal ideation — and they communicate in English, Hindi, and Hinglish. The system runs as a stateful FastAPI container on Railway with a React/Vite frontend.

The product promise is not a smarter chatbot. It is the feeling of being genuinely remembered — and the experience of being understood in this moment. Every architectural decision in this document serves those two outcomes.

NORTH
STAR

Would a perceptive, caring person who genuinely remembered this user do this? If yes — implement it. If not — remove it, no matter how technically elegant it is.

1.1 Architecture Principles

- **One architecture. No feature flags. No dual-mode logic. No legacy paths. No fallback pipelines.**
- Emotional state is modeled continuously, not classified once per message.
- Intervention selection is deterministic (rule-based), not probabilistic (LLM-decided).
- Memory feels like a friend remembering — not a database querying your file.
- Safety is layered: deterministic regex → cognitive layer rules → output auditor.
- Every write is async. Every read is on the hot path and must complete in <800ms.

1.2 System Components at a Glance

System	Component	File	Role
COMPASS	CrisisManager	app/crisis/crisis_manager.py	Layer 1 safety: deterministic keyword sentinel
COMPASS	EmotionalArcReader	app/core/emotional_arc.py	Pure-Python within-session valence trajectory
MEMOIR	MemoryRetriever	app/agents/memory_retriever.py	MEMOIR-scored top-7 memories, 3-thread parallel
COMPASS	CognitiveLayer	app/core/cognitive_layer.py	Single Groq call → intent, risk, intervention, style
COMPASS	CognitivLayerOutput	app/core/cognitive_layer_types.py	Typed dataclass, 14 cl_* ctx keys

COMPASS	PipelineOrchestrator	app/pipeline/pipeline_orchestrator.py	Route dispatch, context assembly, lifecycle
COMPASS	ResponseGenerator	app/agents/response_agent.py	_build_system_prompt_v2, Azure/GLM stream
COMPASS	crisis_templates	app/core/crisis_templates.py	Static warm crisis responses, 4 languages
COMPASS	OutputSafetyAuditor	app/core/output_safety_auditor.py	Post-stream harm pattern detection
COMPASS	EmotionalArcUpdater	app/core/emotional_arc_updater.py	Logs arc delta post-stream
MEMOIR	MemoryStore	app/agents/memory_store.py	add_structured: SignalClassifier→Extractor→QG→CRUD
MEMOIR	MemoryCRUD	app/core/memory_crud.py	Qdrant + Supabase read/write
MEMOIR	MEMOIRScorer	app/core/memoir_scorer.py	6-dimension composite scoring
MEMOIR	MemorySuppressor	app/core/memory_suppressor.py	Hard retrieval filters
MEMOIR	ContextComposer	app/core/context_composer.py	Formats top-7 memories into prompt block
MEMOIR	DecayEngine	app/core/decay_engine.py	Nightly decay, archive, soft-delete
MEMOIR	SessionLifecycle	app/core/session_lifecycle.py	on_session_start/on_message/on_session_end
MEMOIR	MemoryManager	app/agents/memory_manager.py	Public facade for all memory operations

Part 2 — COMPASS: Response Generation Architecture

2. COMPASS — Response Generation

2.1 The Core Shift

The legacy system asked: 'What kind of message is this?' and routed to a fixed handler. A user who begins casually and ends with suicidal ideation was classified 'casual' and received 1–3 cheerful sentences. This is a clinical failure mode, not an edge case.

COMPASS asks: 'What is this person's emotional state right now, how did they get here, and what does a caring human do next?' Emotional state is re-evaluated on every message. Intervention selection is deterministic, not probabilistic.

RESEARCH

Research basis: Limbic's cognitive layer (Nature Medicine, 2026) — augmenting LLMs with deterministic expert reasoning outperformed both standalone frontier LLMs and licensed human therapists on validated CBT competency scales. The cognitive layer did the heavy lifting, not the base model.

2.2 Request Lifecycle — 8 Stages

Every user message flows through 8 stages. Stages 0–2 run sequentially. Stage 1 sub-stages run in parallel within stage 1. The LLM stream begins at Stage 4. Stages 6A–6D run post-stream in daemon threads and never block the response.

Stage	Component	Target latency	Blocks stream?
0	JWT validation · rate limit · session resolve	~10ms	Yes
1A — parallel	Crisis Sentinel (CrisisManager.check_crisis_keywords)	~1ms	Yes
1B — parallel	MEMOIR Memory Retrieval (MEMOIR-scored top-7)	~80ms	Yes
1C — parallel	Emotional Arc Reader (pure Python, VADER + Hinglish)	~3ms	Yes
2	Cognitive Layer — single Groq/Qwen-32b call	~150ms	Yes
3	Context Assembler (ContextComposer + _build_system_prompt_v2)	~10ms	Yes
4	ResponseGenerator — Azure/GLM stream=True	~400ms to first token	Yes

5	SSE stream to client	Ongoing	No — stream is live
6A — post-stream	Memory write: SignalClassifier → MemoryExtractor → QualityGate → CRUD	Async	No
6B — post-stream	Output Safety Auditor (regex harm scan on generated text)	Async	No
6C — post-stream	Emotional Arc Updater (logs arc delta for evaluation)	Async	No
6D — post-stream	SessionLifecycle.on_message() (message_count++, checkpoints)	Async	No

LATENCY	First-token target: <800ms p95. Achieved by running crisis sentinel, memory retrieval, and arc reader concurrently in Stage 1, combined with one cognitive layer call replacing two legacy sequential LLM calls.
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2.3 Cognitive Layer — Full Specification

The Cognitive Layer is the architectural heart of COMPASS. One Groq/Qwen-32b call per message. It replaces IntentRouter, both AnalysisEngine paths (Path B and C), and the emotional trend LLM call. Output is a typed CognitivLayerOutput with 14 fields.

2.3.1 Inputs

Input	Source	Used for
user_message	Current HTTP request	Primary text for analysis
recent_turns	Last 6 turns from session buffer	Conversation context for LLM
arc	EmotionalArcReader.compute_arc() — Stage 1C	Trajectory context, risk escalation
crisis_sentinel_level	CrisisManager.check_crisis_keywords() — Stage 1A	Hard/ambiguous → override rules
session_count	session_registry in Supabase	Relationship gate for memory injection

2.3.2 LLM Call Parameters

- Model: Groq/Qwen-32b — same groq_nlp.client instance, no new connections
- Temperature: 0.1 — deliberately low for deterministic structured output
- Max tokens: 256 — output is small JSON only
- Timeout: 8.0 seconds — fallback activates on any failure
- Hard crisis skip: when crisis_sentinel_level == 'hard', LLM call is skipped entirely. Crisis is already confirmed by deterministic sentinel.

2.3.3 LLM Output Schema

```
{
  "intent": "venting | advice | casual | reflect | update | crisis",
  "primary_emotion": "one word — anxious | sad | overwhelmed | hopeful | ...",
  "emotional_valence": float // -1.0 (very negative) to 1.0 (very positive)
  "emotional_intensity": float // 0.0 (mild) to 1.0 (very strong)
  "risk_level": "low | moderate | elevated | crisis",
  "language_mirror": "en | hi | hinglish",
  "cultural_context": "brief note if cultural pressure present, else empty",
  "confidence": float // 0.0 to 1.0
}
```

2.3.4 Deterministic Override Rules (always applied after LLM output)

1. Crisis sentinel hard → force intent='crisis', risk_level='crisis'. LLM output discarded for these fields.
2. Ambiguous sentinel + LLM returned risk_level='low' → escalate to 'moderate'. Never leave ambiguous as low.
3. Arc falling sharply (arc_delta < -0.4) → escalate risk one level. Low→moderate, moderate→elevated.

4. Any LLM call failure or parse failure → fallback: intent='emotional', risk_level='moderate'. Never default to 'low'.
5. fallback_used=True set in CognitivLayerOutput on any failure. Logged for monitoring.

2.3.5 Deterministic Downstream Derivations

These fields are computed from the LLM output via pure Python rule maps — not additional LLM calls.

Output field	Source	Key logic
intervention_sequence	INTERVENTION_RULES dict — (risk_level, intent) → list	Ordered 1–3 micro-interventions: validate, reflect, ground, reframe, affirm, explore, practical_support
mi_move	MI_MOVE_MAP dict — intent → OARS technique	venting→reflection, advice→open_question, casual→no_move, crisis→reflection
response_length	RESPONSE_LENGTH_MAP dict — intent → length	casual→short, venting→medium, reflect→long, crisis→medium
question_allowed	Compound rule	False if intent==venting OR crisis OR (arc falling AND intensity>0.7). True otherwise.

2.3.6 ctx Keys Injected (14 cl_* fields)

ctx key	Type	Consumed by
cl_intent	str	route_and_execute() dispatch, MEMOIR I-term
cl_primary_emotion	str	Prompt v2, psychological_analysis dict
cl_emotional_valence	float	MEMOIR E-term scoring, arc updater
cl_emotional_intensity	float	question_allowed rule, prompt v2 intensity label
cl_arc_trajectory	str	Prompt v2 arc note, _build_intervention_directive()
cl_arc_delta	float	Eval trace, risk escalation rule
cl_risk_level	str	Crisis gate, sensitive memory suppressor, prompt safety note
cl_intervention_sequence	list[str]	_build_intervention_directive()
cl_response_length	str	Prompt v2 length guidance
cl_question_allowed	bool	enforce_question_budget(), prompt v2, ContextComposer gate
cl_language_mirror	str	Prompt v2 language guidance, directive builder
cl_mi_move	str	Prompt v2 MI guidance
cl_cultural_context	str	Prompt v2, psychological_analysis dict
cl_fallback_used	bool	Eval trace, [COMPASS] monitoring logs

2.4 Emotional Arc Reader

Pure Python. No LLM. No external API. Runs in ~3ms as Stage 1C, parallel with memory retrieval and the crisis sentinel. Tracks the user's within-session emotional trajectory.

2.4.1 Valence Scoring (per user message)

6. VADER SentimentIntensityAnalyzer compound score → `base_score` (-1.0 to 1.0).
7. Hinglish negative markers scan (examples: 'nahi', 'akela', 'dard', 'koi nahi sunta', 'thak gaya', 'ro raha', 'bechaini', 'takleef'). Each hit → -0.15, capped at -0.45 total.
8. Hinglish positive markers scan (examples: 'theek hun', 'khush', 'better feel', 'acha lag', 'shukriya'). Each hit → +0.15, capped at +0.45 total.
9. `final_score` = `clamp(base_score + hinglish_delta, -1.0, 1.0)`. Rounded to 3dp.
10. VADER unavailable fallback: use Hinglish adjustment only (`base_score` = 0.0). Arc reader never throws.

2.4.2 Arc Direction (window of last 8 user messages)

Direction	Condition
rising	<code>recent_avg</code> (last 3 scores) > <code>earlier_avg</code> by ≥ 0.1
falling	<code>recent_avg</code> (last 3 scores) < <code>earlier_avg</code> by ≥ 0.1
volatile	<code>max(scores) - min(scores) > 0.6</code> AND both positive and negative scores present in window
stable	<code>abs(delta) < 0.1</code> OR fewer than 3 user messages scored

2.4.3 Arc Output Fields

- `current_valence` — score of the most recent user message
- `arc_direction` — rising | falling | stable | volatile
- `arc_delta` — `scores[-1] - scores[-4]`, rounded to 3dp (magnitude of recent change)
- `session_low` — minimum valence seen in this session's window
- `session_high` — maximum valence seen in this session's window

2.5 Route Dispatch & Path Handlers

`route_and_execute()` in `PipelineOrchestrator` reads `cl_intent` and `cl_risk_level` from `ctx` (populated by `CognitiveLayer`) and dispatches to the correct handler. There is no `IntentRouter`. There is no `AnalysisEngine`. Those are removed.

Condition	Path	What happens
<code>cl_risk_level=='crisis'</code> OR <code>cl_intent=='crisis'</code>	D-crisis-warm	Warm template selected by language+severity.

		response_gen.generate() never called. ctx['ai_response'] set directly. ctx['response_generated']=True.
cl_intent=='casual'	A-casual	_path_light() — no dummy data. Real cl_risk_level used. Minimal psychological_analysis dict built from cl_* fields.
cl_intent in ('venting','emotional','reflect','update')	B-emotional	_path_standard() — no AnalysisEngine call. psychological_analysis built from cl_* fields. intervention_directive from _build_intervention_directive().
cl_intent=='advice' or else	C-therapeutic	_path_rich() — no AnalysisEngine call. Same cl_* field mapping. Deeper intervention_sequence used.

SAFETY

Eliminated: dummy {risk_assessment: 'low'} injection in _path_light(). A casual-classified message now carries real cl_risk_level from the cognitive layer. Subtle suicidal disclosures in 'casual' messages are no longer invisible.

2.6 System Prompt V2

RESPONSE_SYSTEM_PROMPT_V2 in app/core/prompts.py is the sole active prompt template. It has 12 injectable variables. _build_system_prompt_v2() in ResponseGenerator assembles it on every request.

2.6.1 Prompt Sections and Variables

Section	Variable(s)	Source
Identity	companion_name	companion config — static
Relationship	stage_directive	_get_stage_directive(stage) — trust window / deepening / insight
Memory context	memory_context	ContextComposer output — MEMOIR pipeline
This moment	primary_emotion, emotional_intensity_label, arc_trajectory, arc_note	cl_primary_emotion, cl_emotional_intensity, cl_arc_trajectory
Task this turn	intervention_directive	_build_intervention_directive(cl_intervention_sequence)
How to respond	mi_guidance, language_guidance, length_guidance	cl_mi_move, cl_language_mirror, cl_response_length
Safety	safety_note	Injected when cl_risk_level is 'elevated' or 'crisis'. Empty when 'low'/'moderate'.
Remember	personality_instruction	companion config — static

2.6.2 Intervention Directive Format

`_build_intervention_directive()` converts `cl_intervention_sequence` (`['validate','reflect','ground']`) into a formatted multi-line instruction string:

```
Step 1: Acknowledge what the user is feeling without judgment. Name the emotion if clear.
Step 2: Use reflective listening – mirror the essence, not the words.
Step 3: Gently help the user connect to the present.
[when cl_question_allowed=False] Do NOT ask any question this turn.
[when cl_language_mirror='hinglish'] Mirror their Hinglish naturally.
[when arc='falling'] Note: user's emotional state is declining. Warmth over advice.
```

2.6.3 Total Prompt Token Budget

Section	Max tokens	Notes
Identity + persona instructions	~80	Static, always present
Stage directive	~60	Changes by session tier
Memory context (ContextComposer)	550	Hard cap enforced by ContextComposer
Cognitive layer context (emotion, arc, task, guidance, safety)	~150	Derived from <code>cl_*</code> fields
Personality instruction	~40	Static
Total system prompt	~880	Well within model context limits

2.7 Path D — Warm Crisis Response

When `cl_risk_level=='crisis'` or `cl_intent=='crisis'`, the system returns a static warm template. Static = deterministic = safe. The templates maintain the companion persona voice while embedding safety content. `response_gen.generate()` is never called on this path.

2.7.1 Template Selection

- Language detected from `ctx['language_preference']` or `ctx['content_locale']`
- Severity: 'hard' (confirmed crisis via keyword sentinel) or 'elevated' (ambiguous/elevated risk from cognitive layer)
- 4 languages × 2 severities = 8 static templates. Every template contains: iCall 9152987821 + Vandrevale 1860-2662-345
- Language fallback: unrecognized language → Hinglish template

2.7.2 Template Voice Principles

- Maintains companion name and persona — never drops to clinical language
- Acknowledges the disclosure with warmth first, safety content second
- Hard template: 'Yaar, I'm really glad you said something right now...' [Hinglish example]
- Elevated template: softer check-in that invites the user to say more while embedding hotlines

2.8 Post-Stream: Output Safety Auditor

After the SSE stream completes, `OutputSafetyAuditor.run_async()` launches in a daemon thread. Rule-based only. Never blocks the stream. This is a backstop — the primary safety layers are the crisis sentinel and cognitive layer `risk_level`.

Pattern	What it catches	Severity	Effect
<code>self_harm_method</code>	Specific self-harm method mentions in bot response	critical	<code>passed=False</code> , <code>logger.error</code> with <code>session_id</code>
<code>specific_method_detail</code>	Lethal dose, 'how to die', etc.	critical	<code>passed=False</code> , <code>logger.error</code> with <code>session_id</code>
<code>dismissive_minimization</code>	'Just cheer up', 'stop being', 'snap out of'	warning	<code>passed=True</code> , <code>logger.warning</code>
<code>unsolicited_diagnosis</code>	'You have depression', 'you suffer from anxiety'	warning	<code>passed=True</code> , <code>logger.warning</code>
<code>false_safety_claim</code>	'I can always help', 'I will never leave you'	warning	<code>passed=True</code> , <code>logger.warning</code>

- All exceptions inside `run_async()` caught and logged — never propagated
- Patterns compiled at class instantiation, not per-call

- Logs [SAFETY-AUDIT] prefix for all events — filterable in Railway logs

2.9 LLM Budget per Request

Call	Model	Purpose	~Latency	~Cost
1	Groq/Qwen-32b	Cognitive Layer: intent + emotion + risk + all intervention selection	150ms	\$0.0003
2	Azure/GLM stream	ResponseGenerator — full response generation	400ms→1st token	\$0.003
—	None	EmotionalArcReader (pure Python + VADER)	3ms	\$0
—	None	CrisisManager keyword sentinel (regex)	1ms	\$0
—	None	OutputSafetyAuditor (regex, async post-stream)	async	\$0

Total on critical path: 2 LLM calls. No sequential analysis calls. First-token delivery target: <800ms p95.

Part 3 — MEMOIR: Memory System

3. MEMOIR — Memory Architecture

MEMOIR is the memory layer that makes MindMitra feel like it genuinely knows you. It stores five types of structured, typed memories, retrieves them using a 6-dimension composite score, and injects them as warm natural language — not as a database readout.

3.1 Memory Type Taxonomy

Five distinct types serve different psychological functions. Mixing them without type-awareness produces incoherent, context-blind responses. Each type has a different extraction trigger, decay rate, and retrieval priority.

Type	What it stores	Examples	Retention	Decay λ
identity	Stable facts about who the person is. Changes rarely — only on explicit statement.	Name, age, city, occupation, family structure, pronouns, language preference	2 years from last_confirmed	0.001 (very slow)
preference	How they like to be treated. More mutable than identity — shifts over months.	Prefers direct advice over validation · dislikes therapy-speak · finds breathing exercises useful · journals mornings	1 year, soft decay after 6mo	0.002 (slow)
behavioral	Observed patterns across sessions. What they actually do, not what they say about themselves.	Messages late at night during stress · session length increases when discussing family · abandons conversation when asked about medication	180 days — behaviors shift fast	0.005
emotional	Emotional history across sessions. The most sensitive category. Drives felt understanding.	Father passed in March · panic attacks before presentations · fear of failure · got the promotion they were anxious about for weeks	1yr acute, 5yr for grief/loss, permanent for crisis	0.004
contextual	Ongoing situations being navigated. Time-bounded — stale once resolved.	Preparing for interview next week · going through divorce · starting therapy · waiting for medical results	90 days from last update	0.008 (fast)

POLICY

Retention policy: MindMitra never permanently deletes user memories without explicit user request. 'Forgetting' means archiving — moving to a non-retrievable state while preserving the audit trail. Crisis memories are never auto-archived and require manual review.

3.2 Storage Architecture — Three Layers

3.2.1 Layer 1 — Vector Store (Qdrant)

Semantic search backbone. Holds embeddings for all memory content. The only layer that answers 'find memories semantically related to this query.'

Field	Details
Embedding model	BAAI/bge-m3 — 1024 dimensions. Natively multilingual. Handles English, Hindi, and Hinglish code-switching. Replaces all-MiniLM-L6-v2.
Query prefix	embed(text, is_query=True) prepends BGE-M3 retrieval prefix. Document embeddings use is_query=False. Critical for retrieval quality.
Collection	companion_memories — single collection, user isolation via user_id payload filter on every query
Index	HNSW with ef=128 for accuracy/speed tradeoff. Cosine similarity distance metric.
Payload per vector	memory_id, user_id, type, content, verbatim_anchor, session_id, created_at, last_accessed, access_count, confidence, emotional_valence, emotional_intensity, tags[], decay_score, is_sensitive, is_active, is_resolved, supersedes_id, importance_score, language

3.2.2 Layer 2 — Relational Store (Supabase / PostgreSQL)

Source of truth for all structured metadata, scoring, retention, and audit trail. Vector search finds candidates; Supabase provides the ground truth for all metadata operations.

Table	Purpose	Key indexes
memory_metadata	Full mirror of Qdrant payload for SQL querying, filtering, and contradiction detection	(user_id, type), (user_id, importance_score DESC), (user_id, last_accessed DESC)
memory_contradictions	Flagged conflicts: memory_id_a, memory_id_b, detected_at, status, resolution_memory_id	(user_id, status)
session_registry	One row per session: started_at, ended_at, message_count, summary_written, narrative_updated	(user_id, started_at DESC)
user_memory_profile	session_count, trust_tier (1–5), language_preference, last_session_at, narrative_paragraph	(user_id) — single row per user
session_summaries	Gemini-generated session summaries: themes, emotional_arc, key_disclosures	(user_id, created_at DESC)

- All tables have RLS policies: users can only read/write their own rows (user_id = auth.uid())
- Write pattern: append-only + soft delete (is_active=False). No hard deletes except on explicit user request.

3.2.3 Layer 3 — Session Cache (Redis)

Hot-path cache for active sessions. Prevents redundant DB calls within a session.

Redis key	TTL	Contains
user:{id}:memory_context	600s — refreshed on retrieval	Assembled ContextComposer output for current session
user:{id}:has_memories	120s — short because new users gain memories fast	Boolean: skip retrieval entirely for new users
user:{id}:session_buffer	session duration + 30min	Last 20 turns for arc reader and cognitive layer context

NOTE	Redis is required before scaling beyond a single worker. session_message_counters is currently an in-memory Python dict — a known concurrency bug under multi-worker deployment. Migrate to Redis INCR before scaling. Until Redis is available, use Supabase messages table for the session buffer.
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3.3 Memory Creation Pipeline

Memory extraction runs post-stream in a daemon thread. It never blocks the user-facing response. The write path has 4 sequential stages.

3.3.1 Stage 1 — SignalClassifier (rule-based, <5ms)

Determines if a message is memory-worthy. No LLM. Fires on:

- First-person statements: 'I am', 'I've been', 'My X is', 'mera', 'mujhe', 'main'
- Named entities: capitalized proper nouns in personal context, explicit person names
- Emotional intensity markers: 'never', 'always', 'devastated', 'breakdown', 'koi nahi sunta'
- Explicit disclosures: 'I haven't told anyone', 'honestly', 'truth is'
- Resolution statements: 'I got the job', 'we broke up', 'finally feeling better'
- Any message with intensity > 0.7 (from arc reader) triggers extraction regardless of other signals — hot-path extraction for high-emotional-intensity messages
- Crisis language: always triggers, regardless of interval counter

3.3.2 Stage 2 — MemoryExtractor (single LLM call)

Model: claude-haiku-4-5 (fast, conservative, structured output). Extracts typed memory candidates with a verbatim_anchor requirement.

Key extraction rules enforced in prompt:

- Only extract what the user explicitly stated. No inference.
- Every extracted memory must include a verbatim_anchor: the exact user phrase that justifies it. No anchor = candidate discarded.
- Maximum 6 candidates per call. Prioritize the most significant.
- confidence < 0.5 on any extraction where the quote is not direct: cap at 0.5, not 0.9
- Hypotheticals not stored ('If I had a better job'). Questions not stored as facts. Reported speech flagged.

3.3.3 Stage 3 — QualityGate (pure Python)

Hard filters before any write:

- confidence < 0.45 → rejected
- verbatim_anchor missing or empty → rejected
- Injection attack sanitizer: strips XML/HTML tags, prompt injection patterns ('ignore previous', 'system:', 'SYSTEM', 'disregard', 'forget everything'). Rejected + logged if found.
- Deduplication: embed candidate, query Qdrant for top-1 similar for this user. cosine > 0.91 → reinforce existing (access_count++) rather than insert new. 0.75–0.91 → flag as possible_duplicate.
- Contradiction detection: new semantic memory flagged against top-3 similar existing → written to memory_contradictions if conflict detected.
- Safety escalation: is_sensitive=True AND intensity > 0.8 → confidence floor = 1.0. Never drop high-intensity sensitive memories on confidence grounds.

3.3.4 Stage 4 — MemoryCRUD (write)

- `insert()`: embed with BGE-M3 (`is_query=False`) → Qdrant write + Supabase `memory_metadata` write. Returns `memory_id`.
- `reinforce()`: Qdrant `set_payload` `access_count++` and `last_accessed=now()`. Supabase update.
- `soft_delete()`: Qdrant `set_payload` `is_active=False`. Supabase update with reason. No physical delete.
- `supersede()`: insert new → set old `is_active=False` with `supersedes_id` pointer.
- `log_contradiction()`: insert row into `memory_contradictions` with `status='pending'`.

3.3.5 Extraction Trigger Schedule

Trigger	What runs
Any message with intensity > 0.7	Hot-path immediate extraction: SignalClassifier → MemoryExtractor → QualityGate → CRUD
Every 12 messages (MEMORY_TRIGGER_INTERVAL)	Full extraction pass on last 12 messages
Every 36 messages	Full extraction + <code>generate_session_summary()</code> via Gemini Flash
/chat/end-session endpoint	Final extraction + session summary + <code>session_registry.ended_at</code>
Every 10 sessions (<code>session_count % 10 == 0</code>)	Narrative update: Gemini synthesizes 200–250 word user narrative → <code>user_memory_profile.narrative_paragraph</code>
Nightly cron (DecayEngine.schedule_nightly_decay)	Recompute decay scores. Archive <0.10. Soft-delete <0.05.

3.4 Composite Scoring Formula

The composite score determines which memories surface in retrieval. Computed fresh at retrieval time — never pre-computed. This ensures recency and emotional state always reflect the current moment.

3.4.1 MEMOIR Score — 6 Dimensions

Dim	Signal	Formula	Weight
M	Meaning relevance	$\text{cosine_similarity}(\text{query_embed}, \text{memory_embed})$ — BGE-M3 vectors, $\text{is_query}=\text{True}$ for query	0.25
E	Emotional congruence	$1.0 - \text{memory_valence} - \text{current_valence} \times \text{current_intensity}$. Source: $\text{cl_emotional_valence} + \text{cl_emotional_intensity}$ from cognitive layer.	0.20
M	Momentum (recency)	$\exp(-\lambda \times \text{days_since_accessed})$. λ per type: identity=0.001, preference=0.002, behavioral=0.005, emotional=0.004, contextual=0.008	0.15
O	Observational reinforcement	$\min(\text{access_count} / 8.0, 1.0)$ — memories reinforced through repeated access are more stable (Nader 2000)	0.15
I	Intent alignment	(intent, type) matrix → multiplier 0.5–1.5, normalized. venting+emotional=1.5, advice+behavioral=1.4, casual+identity=1.2	0.15
R	Relationship gate	session_count 1–3: sensitive memories × 0.3. session_count 4–8: × 0.7. session_count 9+: × 1.0	0.10

Final score: $S = 0.25 \cdot M(\text{meaning}) + 0.20 \cdot E + 0.15 \cdot M(\text{momentum}) + 0.15 \cdot O + 0.15 \cdot I + 0.10 \cdot R$

Hard floor: $S < 0.25 \rightarrow$ dropped before injection. No memory below this threshold enters the prompt.

3.4.2 Category Importance Overrides (applied before scoring)

Memory type / subtype	Importance floor	Effect
type = 'emotional', subtype = 'crisis'	10.0 (forced)	Always surfaces in crisis + therapeutic intents
type = 'behavioral' (coping that worked)	8.0 minimum	Ensures effective coping strategies always retrievable
subtype = 'milestone' (major life event)	7.0 minimum	Major moments never lost to low importance
subtype = 'win' (positive outcome)	6.0 minimum	Positive references maintained — not just struggles

3.4.3 Resolved Emotional Memories

Emotional memories with `resolution_status = 'resolved'` have their intensity halved for scoring:
 $E_{\text{effective}} = \text{intensity} \times 0.5 \times (1 + 0.3 \text{ if unresolved})$. This prevents old resolved grief from overwhelming new signals.

3.5 Retrieval Pipeline — 5 Stages

Stage 0 — Short-Circuit Check (<1ms)

Query has_`_memories` Redis key (120s TTL). If False: skip all retrieval, return empty. New users pay zero retrieval cost. Cache miss → COUNT query on Supabase `memory_metadata`.

Stage 1 — Query Understanding (<50ms)

- Embed user_message with BGE-M3 (`is_query=True`)
- Expand emotional vocabulary: 'stressed' → 'stressed, anxious, overwhelmed, tense' (improves semantic recall)
- Prepend intent label to query: 'emotional context: [message]' — steers embedding toward emotional semantic space
- Extract named entities: capitalized proper nouns boosted in keyword search thread
- Hinglish normalization: common romanized Hindi → transliterated equivalents for better embedding alignment

Stage 2 — Parallel Candidate Retrieval (3 threads, <80ms total)

- Thread A — Dense vector search: Qdrant top-25 by cosine similarity, filtered `user_id + is_active=True`
- Thread B — Keyword/structured search: Supabase `memory_metadata` WHERE tags @> named_entities OR type='behavioral', limit 15
- Thread C — Recency anchor: Supabase `memory_metadata` ORDER BY last_accessed DESC LIMIT 5 — always includes recently accessed memories regardless of semantic match
- Merge and deduplicate by `memory_id` → ~35–45 candidates. Hard timeout: 7 seconds. On Qdrant timeout: proceed with empty (chat continues). On Supabase metadata timeout: proceed with default importance scores.

Stage 3 — MemorySuppressor (hard filters, <10ms)

- `is_active=False` → suppress
- `decay_score < 0.08` → suppress
- `confidence < 0.35` → suppress
- `is_sensitive=True` AND `emotional_intensity > 0.85` AND `intent != 'crisis'` AND `session_count < 6` → suppress (early session, sensitive memory)
- `type='emotional'`, `is_resolved=True`, user not explicitly referencing → suppress
- Crisis override: when `cl_intent='crisis'`, NEVER suppress sensitive memories. Override rules 4 and 5.

Stage 4 — MEMOIR Scoring

Each surviving candidate receives its S score using the formula in §3.4.1. Crisis memories: if `is_sensitive=True` AND `intent='crisis'` → S = 1.0 (hard-pinned to top). Sort by S descending.

Stage 5 — Top-k Selection and Reinforcement

- Select top 7 memories
- Type diversity: max 3 memories of the same type in top-7. Lowest-scoring excess replaced by next-highest of different type.
- Intent-aware bucket allocation (caps per intent):

Bucket	Casual	Emotional	Therapeutic	Crisis
Identity + contextual + preference	3	5	7	4
Behavioral (coping patterns)	1	2	4	3
Emotional (struggles, wins, fears)	1	2	3	4
Reflective (session patterns)	0	1	2	1

- Reinforce all selected memories: `MemoryCRUD.reinforce()` → `access_count++`, `last_accessed=now()` in background thread.

3.6 Memory Injection — ContextComposer

ContextComposer formats the top-7 memories into a structured natural-language briefing. This is what makes the AI feel like it knows the user — not a database readout, but a warm internal briefing.

3.6.1 Injection Format

```
WHAT YOU KNOW ABOUT THIS PERSON

About them: [Name] is a [age-bracket] [occupation] based in [location].
           They prefer [communication_style].

What they're going through: [active contextual memories, most important first]

What has helped them before: [behavioral memories — coping that worked]

Things you remember them sharing: [top emotional/identity memories by composite
score]

Patterns you've noticed: [behavioral + reflective memories, if session_count > 2]

Recent emotional trend: [one sentence from emotional arc + session summary synthesis]
```

3.6.2 Section Priority and Token Budget

Section	Max tokens	Always included?
Identity + preference summary	150	Yes — never truncated
Active contextual memories	250	Yes if any exist
Emotional + behavioral memories	250	Yes if any exist
Reflective patterns	100	Only if session_count > 3
Emotional trend line	50	Always
Total hard cap	550	Enforced by ContextComposer. Truncate from lowest composite score up.

3.6.3 Narrative Mode (session ≥ 15)

For established users with ≥15 sessions, ContextComposer switches to narrative mode: the user_memory_profile.narrative_paragraph (200–250 words synthesized by Gemini every 10 sessions) replaces the bullet-point memory list. The model sounds like it truly knows the person — not like it is recalling facts.

3.6.4 Injection Gate: cl_question_allowed

When `cl_question_allowed=False` (venting or crisis), ContextComposer suppresses episodic and affective memory injection. Only procedural (how to engage) and relational context are included. This prevents surfacing painful memories when the user just needs to be heard.

3.6.5 Sanitization Before Injection

Every memory bullet passes through `sanitize_for_injection()` before assembly:

- Strips XML/HTML tags: regex `<[>]+>`
- Strips injection patterns: 'ignore previous instructions', 'system:', 'you are now', 'forget everything', 'disregard', 'new instructions', '<system>'
- Strips control characters and null bytes
- Truncates any bullet exceeding 200 characters (ellipsis)
- Strips URLs and file paths
- Pattern matching is case-insensitive. Filtered bullets replaced with '[content filtered]'. Logged with `memory_id`.

3.7 Memory Evolution & Decay

3.7.1 Update Logic

Scenario	Action
New value, same field, explicit user statement	supersede(): insert new, set old is_active=False with pointer
New value, same field, inferred	Add as candidate, increment frequency_count if pattern repeats
Contradicts existing high-confidence (>0.8)	log_contradiction() → memory_contradictions table. Keep both. Resolve on next session confirmation.
Additive info (more detail on existing fact)	Update content field, refresh last_confirmed, keep original memory_id
Status change: user signals resolution	Update is_resolved=True. Create win/milestone emotional memory if positive outcome.
Preference reinforcement	Reinforce(): access_count++, confidence recalculated

3.7.2 Contradiction Resolution Flow

11. Contradiction flagged in QualityGate → memory_contradictions table: status='pending'
12. Next session: model asks a natural clarifying question if context permits ('I think you mentioned X — has that changed?')
13. User response triggers resolution: confirmed new value → supersede old. Old value denied → reinforce old, discard new candidate.
14. Identity contradictions (name, age) require explicit user statement before supersede — no inferred resolution.

3.7.3 Decay System (nightly)

```
decay_score = confidence × exp(-λ × days_since_accessed) × min(access_count / 8.0, 1.0)
```

decay_score	Action
< 0.10	Archived: is_active=False. Excluded from retrieval. Retained in audit log.
< 0.05	Soft-deleted: invisible to all retrieval paths. Retained in Supabase for audit.
0.10–1.0	Active. Decay score updated in Qdrant payload and Supabase memory_metadata.

- Runs as daemon thread started at app startup: DecayEngine.schedule_nightly_decay()
- Crisis memories: never auto-archived. Manual review required.
- User-requested deletion: hard delete from both Qdrant and Supabase. Logged to audit trail.

Part 4 — Human-Like Memory Design

4. Designing for Human-Like Memory

This section defines how the system should behave at the human level — not what data it stores, but how it uses that data to make users feel genuinely remembered, understood, and cared for. Technical correctness without this layer produces a system that feels like a surveillance database.

4.1 Emotional Continuity

Emotional continuity is the quality that makes users feel genuinely remembered across time. It is not about recalling facts — it is about holding emotional context the way a caring person would.

4.1.1 The Reference Principle

PRINCIPLE	A memory should be referenced explicitly only when it is directly relevant to what the user just said. The AI should never open a conversation with 'I remember you mentioned your father passed away.' It should hold that information and surface it only when the current message connects to it — the way a good friend would.
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4.1.2 Natural vs Surveillance Reference

Situation	Bad (surveillance)	Good (natural)
User: 'I've been feeling really low lately.' System has grief memory.	'I remember you told me your father passed away in March. How are you coping with that grief?'	'That kind of weight can build up — especially when you're still carrying something as big as losing your dad.'
User starts session after being away for a week.	'According to my records, you last spoke on [date] about your job interview.'	'How did that presentation go? You were pretty stressed about it last time.'
User mentions feeling isolated.	'I have noted that you often feel lonely. This is pattern number 3 in your behavioral profile.'	'It sounds like that loneliness has been sitting with you for a while now.'

4.1.3 Session Continuity Bridge

At the start of each session (first 1–2 messages), the AI may make one natural callback to an active contextual memory — only if composite score > 0.70 AND the last session was within 14 days. Maximum: one bridge per session. Never for crisis or grief history.

Pattern: 'Last time we talked, you were [ongoing situation]. How did that go?' — feels like a friend picking up a conversation.

4.2 Relationship Stages

The familiarity and depth of the AI's engagement should evolve naturally with the relationship. Over-familiarity on session 1 feels invasive. Appropriate distance on session 20 feels cold.

Stage	Sessions	AI behavior	Memory usage
Acquaintance	1–2	Warm but neutral. Asks open questions. Does not assume. Does not name-drop.	Only explicit onboarding data. No inference. No behavioral patterns.
Familiar	3–7	Starts making gentle connections. Light callbacks. Uses name occasionally.	Identity + top preference memories. Light contextual memories.
Trusted	8–20	Natural references to past. Notices patterns gently. Adapts tone without announcing it.	Full retrieval pipeline. Pattern references. Emotional continuity.
Established	20+	Deep personalization. Narrative mode injection. References growth over time.	Full retrieval + narrative paragraph + reflective memories + trend analysis.

The R-term in the MEMOIR score (Relationship gate) enforces this: sensitive memories score 0.3× in sessions 1–3, 0.7× in sessions 4–8, and 1.0× after session 9. The system literally cannot over-share early.

4.2.1 Avoiding Over-Familiarity

- Never open a session with crisis or heavy emotional history
- Never use the user's name more than twice per session
- Never surface memories older than 90 days without $S > 0.80$
- Never reference behavioral patterns explicitly: say 'it sounds like you've been under a lot of pressure' not 'I've noticed you usually message late at night when you're stressed'
- Maximum 2 explicit memory references per response — implicit shaping is preferred over explicit recall

4.3 Tone Personalization

The communication style profile is built from behavioral memory (observed signals across sessions) and injected via the {intervention_directive} and {language_guidance} fields in the prompt. It adapts without being announced.

4.3.1 Communication Style Signals (inferred over sessions)

Signal	Inference method	Effect on response
Prefers direct advice vs wants to vent first	Response to past recommendations: engagement after validation vs advice	intervention_sequence: advice early vs validate first
High vs low formality	User's own message register (formal vocabulary vs casual/slang)	language_guidance: match register
Comfortable with humor vs serious tone	User's emoji use, jokes, playful language	mi_move: no_move allows lighter touch
Prefers short vs detailed responses	Session engagement length, question follow-up rate	response_length: short vs long
Language preference	Detected language per message, stored in user_memory_profile.language_preference	cl_language_mirror: en hi hinglish

4.3.2 Tone Injection Example

```
HOW TO ENGAGE WITH THIS PERSON:
They respond better when given space to process before advice is offered.
They use informal language – match their register, not a formal one.
Keep responses concise – they don't engage well with long paragraphs.
Respond in Hinglish since that's how they write.
```

4.4 What MindMitra Knows — The Full Semantic Profile

A complete semantic profile, built over time across all five memory types, covers four layers. This is the full picture that makes MindMitra feel like it truly knows someone:

Who they are (identity layer)

Name, age bracket, location, occupation, family structure, cultural background, language preference.

How they live (preference + behavioral layer)

When they reach out (timing patterns), what helps them (coping preferences), what frustrates them (negative preferences), how they communicate (style signals), what they're currently navigating (contextual layer).

What they carry (emotional layer)

Their significant struggles, losses, fears, unresolved conflicts, past crises. Their wins, milestones, things they're proud of. Their emotional trajectory across months — improving, staying stuck, or deteriorating.

How they've grown (reflective layer)

Patterns noticed across sessions. Growth observed. Coping strategies that have evolved. Generated by the Gemini session reflection job — not individual facts but the meaning of the facts combined into a narrative.

Part 5 — COMPASS × MEMOIR Integration

5. How COMPASS and MEMOIR Connect

COMPASS and MEMOIR are not two systems running in parallel. They are one system where each layer actively informs the other. Every integration point below is a live data flow, not a conceptual connection.

Integration point	Direction	Data flow
Emotional state → memory retrieval	CognitiveLayer → MEMOIRScorer	cl_emotional_valence + cl_emotional_intensity feed the E-term (emotional congruence). Memories that match the user's current emotional state score higher.
Intent → memory retrieval	CognitiveLayer → MEMOIRScorer	cl_intent feeds the I-term (intent alignment matrix). Venting → emotional memories score higher. Advice → behavioral memories score higher.
question_allowed → memory injection	CognitiveLayer → ContextComposer	cl_question_allowed=False suppresses episodic and affective memory sections. Only procedural + relational injected when user is venting or in crisis.
memory_context → prompt v2	ContextComposer → ResponseGenerator	ContextComposer output is the {memory_context} variable in RESPONSE_SYSTEM_PROMPT_V2. Memory shapes every response.
Risk level → sensitive memory gate	CognitiveLayer → MemorySuppressor	cl_risk_level='crisis' disables MemorySuppressor's sensitive-memory suppression rule. Crisis intent unlocks sensitive memory access.
Arc trajectory → risk escalation	EmotionalArcReader → CognitiveLayer	arc_delta < -0.4 (sharp drop) → cognitive layer escalates risk_level one tier. Arc falling → question_allowed=False.
Session lifecycle → memory write	SessionLifecycle → MemoryStore	on_message() triggers extraction at 12-message intervals. on_session_end() runs final extraction + Gemini summary. Narrative update every 10 sessions.
Narrative mode → prompt v2	user_memory_profile → ContextComposer → ResponseGenerator	user_memory_profile.narrative_paragraph replaces bullet-point memory list for session ≥15. Injected via {memory_context}.
Memory type → intervention selection	MEMOIR retrieval → CognitiveLayer intervention rules	Intent aligned with dominant retrieved memory type influences intervention_sequence selection in INTERVENTION_RULES.

Part 6 — Deployment & Observability

6. Deployment & Observability

6.1 Architecture Mode

Parameter	Value
Architecture	COMPASS + MEMOIR unified — always active. One pipeline. No flags.
Feature flags	None. NEW_PIPELINE_ENABLED and NEW_RETRIEVAL_ENABLED do not exist.
Legacy paths	None. IntentRouter, AnalysisEngine, mem0, and old 4-path pipeline are fully removed.
Rollback strategy	Git revert to previous commit. Deploy previous container image.

6.2 Environment Variables (runtime config only)

Variable	Purpose
GROQ_API_KEY	Cognitive layer Qwen-32b calls
AZURE_OPENAI_KEY / AZURE_ENDPOINT	ResponseGenerator Azure/GLM stream
ANTHROPIC_API_KEY	MemoryExtractor claude-haiku-4-5 calls
SUPABASE_URL + SUPABASE_KEY	Supabase client — memory_metadata, session_registry, all tables
QDRANT_URL + QDRANT_API_KEY	Qdrant vector store
REDIS_URL	Session cache. Required for multi-worker deployment.
ALLOW_EVAL_TRACE	'true' to enable X-MindMitra-Eval-Trace header diagnostics

6.3 Observability — Log Prefixes

Prefix	What it logs	Level
[COMPASS]	Cognitive layer call: intent, risk_level, arc, intervention_sequence, fallback_used	INFO per request

[SAFETY-AUDIT]	OutputSafetyAuditor results: violations[], severity, session_id	ERROR (critical) / WARNING
[ARC-UPDATE]	Post-stream arc delta: user_valence, response_valence, intent, arc_direction	INFO per response
[MEMORY-MEMOIR]	Retrieval: memories_retrieved, types, MEMOIR scores. Write: extracted, approved, rejected, contradictions	INFO per trigger
[SESSION]	Session lifecycle events: start, end, checkpoint, narrative update	INFO
[DECAY]	Nightly decay pass results: processed, archived, soft-deleted per user	INFO daily

6.4 Eval Trace (X-MindMitra-Eval-Trace header)

When ALLOW_EVAL_TRACE=true and header is present, ctx['_eval_data'] is written and returned in response metadata:

- cognitive_layer: {intent, risk_level, arc_trajectory, arc_delta, fallback_used, question_allowed, intervention_sequence}
- arc: {current_valence, arc_direction, session_low}
- memory: {memories_retrieved, types_used, memoir_scores, context_tokens_used}

Part 7 — Failure Modes & Mitigations

7. Failure Modes

Failures in a mental health system carry real risk. A wrong memory retrieval is not a minor UX bug — it can undermine trust, cause distress, or miss a crisis signal. Each failure mode has a defined root cause, detection signal, and mitigation.

Failure	Root cause	Detection	Mitigation
Suicidal user classified as 'casual'	Subtle ideation without hard keywords	Sharp valence drop in arc	1. Arc reader detects drop → risk escalated by deterministic rule. 2. Cognitive layer sees full message + arc context. 3. No dummy data — <code>cl_risk_level</code> is always real.
Cognitive layer LLM failure	Groq timeout or 503	<code>fallback_used=True</code> in <code>eval_trace</code>	Fallback: <code>risk_level='moderate'</code> , <code>intent='emotional'</code> . Never 'low'. Warm response, not crash.
Hinglish retrieval failure	BGE-M3 embedding drift on code-switched text	Non-English users report AI 'forgetting'	BGE-M3 natively multilingual. Hinglish heuristic markers in arc reader. Thread B keyword search catches named entities.
Stale memory retrieved	User's life changed, memory not updated	User contradicts AI reference	Decay archives unconfirmed old memories. Contradiction detection flags. Model uses hedging: 'I think you mentioned...'
Creepy over-personalization	Too many memories, surfaced too early	User feedback or session drop	R-term gates sensitive memories by session count. Max 2 explicit references per response. Behavioral patterns never referenced as patterns.
Prompt injection from memory	Adversarial user crafts extraction bait	[MEMORY-MEMOIR] filtered log	QualityGate strips at extraction. <code>sanitize_for_injection()</code> strips at assembly. Injection patterns case-insensitive.
Crisis memory not persisted	Async thread failure, Supabase timeout	Crisis memory absent in future sessions	Crisis writes use priority thread with 2s timeout. Dead-letter logging. Daily consistency check job.

Missing critical context	High-intensity message before 12-message trigger	User re-discloses in later session	Hot-path extraction: any message with intensity > 0.7 triggers immediate extraction regardless of counter.
OutputSafetyAuditor false negative	Novel harm phrasing not in patterns	Critical log absent for harmful response	Auditor is backstop only. Primary safety: crisis sentinel + cl_risk_level. Patterns expanded from logged events.
Hallucinated memory	Extraction LLM inferred fact not stated	User denies fact: 'I never said that'	verbatim_anchor mandatory. confidence capped at 0.5 without direct quote. User correction triggers supersede.

Part 8 — Scalability

8. Scalability Roadmap

Phase	User scale	Changes required
Current (MVP)	0 → 1,000	Single Qdrant instance · Supabase free tier · Railway single worker · Redis via Upstash (replaces in-memory counter) · Daemon threads for async jobs
Post-beta	1,000 → 10,000	Qdrant horizontal sharding (2–4 shards) · Supabase read replicas · Move extraction + session-end jobs to Celery + Redis queue · APM tracing for all memory pipeline steps
Growth	10,000+	Qdrant Cloud managed cluster with auto-scaling · Supabase Pro with PgBouncer · Separate Celery workers for extraction vs session-end (different resource profiles) · BM25 sparse retrieval added alongside vector search

8.1 Background Job Architecture

Job	Frequency	Current impl	Scale impl
Memory extraction	Every 12 messages + hot-path	Daemon thread	Celery task queue
Session summary (Gemini)	Every 36 messages + /end-session	Daemon thread	Celery + priority queue
Narrative synthesis (Gemini)	Every 10 sessions	Daemon thread	Celery + scheduled task
Decay engine	Nightly	Daemon thread + APScheduler	Celery Beat
Contradiction resolution	Next session after flag	In-request (lightweight)	Same

Part 9 — Research Citations

9. Research Citations

Every significant architectural decision in this document is grounded in published research. This section maps each citation to the specific decision it supports.

Citation	Architectural decision grounded
Rashid et al. (2026). A cognitive layer architecture to support large-language model performance in psychotherapy interactions. Nature Medicine.	Cognitive layer design — deterministic expert rules + small classifier wrapping a general LLM outperformed standalone frontier models AND licensed human therapists on validated CBT competency scales. Grounds the <code>intervention_sequence</code> selection, single-LLM-call approach, and why the cognitive layer (not the base model) does the therapeutic reasoning.
Mukherjee et al. (2024). Polaris: A Safety-focused LLM Constellation Architecture for Healthcare. arXiv:2403.13313.	Parallel fast-lane design: crisis sentinel, memory retrieval, and arc reader running concurrently. Specialist components supervising a primary generative model. Grounds the Stage 1A/1B/1C concurrent architecture.
Hipson & Mohammad (2021). Utterance Emotion Dynamics (UED). ACL 2021.	EmotionalArcReader: valence/arousal trajectory across utterances using the VAD framework. Arc direction classification (rising/falling/stable/volatile) and the <code>arc_delta</code> risk escalation rule.
Miller & Rollnick (2013). Motivational Interviewing, 3rd edition.	OARS framework in <code>mi_move</code> : Open questions, Affirm, Reflect, Summarize. MI spirit = autonomy-preserving, non-directive, empathy-first. Grounds the 'no question when venting' rule (<code>cl_question_allowed=False</code>) and the <code>intervention_sequence</code> ordering (validate before advising).
Ireland & Pennebaker (2010). Language Style Matching.	<code>cl_language_mirror</code> and Hinglish mirroring in responses. Language style synchrony — matching how statements are phrased — is an objective predictor of perceived therapeutic empathy above and beyond the content of reflections.
Bower (1981). Mood-congruent memory retrieval. Psychological Review.	MEMOIR E-term (emotional congruence). Memories matching the user's current emotional valence are more likely to be relevant and to feel natural when referenced. Grounds the 0.20 weight on E.
Tulving (1972, 1985). Episodic and semantic memory. Organization of Memory.	The five memory type taxonomy — episodic and semantic as distinct systems. Extended with behavioral (procedural-like), emotional (affective), and contextual (situational) types for the companion AI context.
Nader (2000). Memory reconsolidation. Nature.	MEMOIR O-term (observational reinforcement). Memories accessed repeatedly become more stable and reliable through reconsolidation. Grounds the <code>access_count</code> weight in scoring.
Ebbinghaus (1885). On Memory (Über das Gedächtnis).	DecayEngine formula: $\exp(-\lambda \times \text{days_since_accessed})$. Type-specific λ values (identity=0.001 slow, contextual=0.008 fast) calibrated to expected stability of each memory category.
Rogers (1965). Client-Centered Therapy.	The MindMitra persona foundation: unconditional positive regard, accurate empathy, and non-directive listening as the necessary

	conditions for change. Grounds the companion persona design ('not a therapist — a friend who listens completely').
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The North Star



Would a perceptive, caring person who genuinely remembered this user do this?

If yes — implement it. If not — remove it.

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